

K878 — RATIFY the shape-forcing (Grace F682: φ = K-invariant Szegő measure on $S^4 \rightarrow$ symmetric vacuum $\rightarrow$ NO free shape knob, only amplitude = α -tower; O6 satisfied) + Elie’s concrete verification (full-rank + shape-matters). BUT the criteria O1–O6 gate only the SYMBOL φ — and the matrix element $M_{ij} = \langle \psi_i | T_\varphi | \psi_j \rangle$ needs TWO inputs: the symbol AND the STATES ψ_i (the three generation K-types). The states are NOT yet gated. If φ is forced but the three K-types are CHOSEN to land on 1:207:3477, the fit just moved from the symbol to the state-labels — a fit dressed as a derivation. New criterion O7: the three generation K-types must be FORCED (anchored at the banked electron $k=1$ + the geometry), not chosen. And the key unification: O7 IS the paper’s PENDING $e/\mu/\tau \rightarrow$ phase assignment — the muon VALUE verdict and the paper’s open phase-assignment are THE SAME question. Resolve once, both close.

Keeper | 2026-07-24 Fri | The last derived-vs-fit gate: gate the STATES, not just the symbol. And it unifies the two open items. Plain.

RATIFIED — the shape is forced (Grace F682, Elie verified)

- Grace F682: φ = the K-invariant Szegő measure on S^4 . The vacuum is K-symmetric $\rightarrow$ NO free shape knob; the only freedom is the overall amplitude = the separate α -tower scale (exactly what O6 allows). $\hat{O} = T_\varphi = P \cdot (\text{Szegő-}S^4 \text{ projection}) \cdot P$. O6 satisfied — shape forced by symmetry.
- Elie verified concretely: on the disk Bergman space, one radial symbol $\rightarrow$ 3 distinct Toeplitz eigenvalues (full-rank, F681 confirmed); and $\varphi=r^2$ vs $\varphi=r^4$ (same locus, different shape) give different masses (O6 is the right gate). Good.

★ THE GAP O1–O6 miss: the matrix element needs the STATES too

$M_{ij} = \langle \psi_i | T_\varphi | \psi_j \rangle$ has TWO inputs: 1. The SYMBOL φ — locus (F583) + shape (F682). FORCED. 2. The STATES ψ_i = the three generation K-types (which $SO(5) \times SO(2)$ harmonics the generations occupy). NOT gated. The eigenvalues depend on BOTH. If φ is fully forced but the three K-types are CHOSEN to hit 1:207:3477, that’s a FIT — the fit just relocated from the symbol to the state-labels. This is exactly Grace’s honest edge (“which harmonic the ν_R condensate occupies,” “the K-types pinned”) — she stopped precisely here and refused to choose the labels (correct).

★ NEW criterion O7 — the STATES (K-types) must be forced

O7: the three generation K-types are anchored at the banked electron ($k=1$, electron-mass proof) + the geometry (the 3 Wallach phases / the $SO(5)$ harmonic ladder), forming a FORCED ladder — NOT chosen to hit the ratios. Concretely: is it $\{k=1,2,3\}$? $\{k=1,3,5\}$? some other forced set? — determined by the phase-assignment and the anchor, not fit. If the K-types are a free choice, the verdict is a FIT even with φ fully forced. O7 is the last derived-vs-fit gate.

★★ The unification: the muon verdict = the paper’s pending phase-assignment

The three generation K-types (O7) ARE the $e/\mu/\tau \rightarrow$ phase assignment that Grace downgraded to PENDING in the paper (localization-vs-coupling, K876 MODERATE-2 / K877). So the two remaining open items across the

whole program are ONE question: - the muon VALUE verdict needs the K-types forced (O7); - the paper's phase-ASSIGNMENT needs the same $e/\mu/\tau \rightarrow$ phase map pinned. Resolve the generation K-type/phase assignment once (target-innocently, anchored at electron $k=1$), and BOTH close — the paper gets its per-generation assignment, and the muon crank gets its states. This is worth stating: it collapses two open fronts into one rep-theory question.

The verdict-state, precisely

- FORCED: operator (Toeplitz), locus (S^4 , F583), shape (Szegő/K-invariant, F682), machinery (Wylers= α , banked). O1–O6.
- PENDING + must-be-FORCED: the 3 generation K-types = the phase assignment. O7. The last input.
- Then binary: turn the Wylers crank $\rightarrow M_{ij} \rightarrow$ diagonalize (Elie harness) $\rightarrow 1:207:3477 + \text{PMNS (DERIVED)}$ or not (structural, stated plainly). Against O1–O7.
- Stopping-rule-safe: resolving the K-types is a rep-theory lookup (the electron-proof's own K-type structure + the harmonic ladder), NOT a 10th mass-mechanism reframe. Grace's lane; not to be fabricated.

Directions

- ★ GRACE (the one remaining lookup): pin the three generation K-types — which $SO(5)$ harmonics, anchored at the banked electron $k=1$. This is O7 AND the paper's phase-assignment — ONE lookup closes both. Source it (the electron-proof K-type + the Wallach-phase ladder), do NOT choose to hit the ratios. Then the Wylers 3×3 is well-defined.
- ★ ELIE: add O7 to the harness (flag if the K-types were a free choice). The crank is: forced φ (F582/F682) + forced K-types (O7) $\rightarrow M_{ij} \rightarrow$ diagonalize.
- ★ LYRA: the paper's per-generation phase assignment closes with O7 — same lookup. Meanwhile fold in the Keeper F676 rewrite ($\rightarrow$ PASS on the count + structure).
- KEEPER: shape-forcing ratified (O6 done); O7 added (states forced — the last gate); the muon verdict and the paper's phase-assignment UNIFIED into one rep-theory lookup.

— Keeper K878, 2026-07-24. RATIFY F682 (φ shape forced: K-invariant Szegő on S^4 , only amplitude= α -tower free; O6 satisfied) + Elie verification (full-rank + shape-matters). GAP: $M_{ij} = \langle \psi_i | T_\varphi | \psi_j \rangle$ needs BOTH symbol (O1–O6, forced) AND STATES ψ_i (the 3 generation K-types, NOT gated) — a chosen K-type set is a FIT relocated from the symbol. Added O7: K-types FORCED (anchored at banked electron $k=1$ + geometry), not chosen. UNIFICATION: O7 = the paper's PENDING $e/\mu/\tau \rightarrow$ phase assignment — the muon VALUE verdict and the paper's phase-assignment are ONE rep-theory lookup; resolve once, both close. Verdict-state: operator/locus/shape/machinery FORCED (O1–O6); K-types the last input (O7); then binary crank $\rightarrow 1:207:3477 + \text{PMNS}$ or structural. See [[Keeper_K877_AUDIT_the_forced_symbol_claim_locus_forced_is_necessary_not_sufficient_07-24]], Grace F682 (Szegő shape), Elie (verification), F583 (locus, electron $k=1$ (banked anchor)).